# Special Theory of Relativity

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# Contents

# 1 Special Theory of Relativity

The special theory of relativity, or simply special relativity, is a scientific theory of the relationship between space and time by Albert Einstein.

## I Postulates

In Albert Einstein's 1905 paper, "On the Electrodynamics of Moving Bodies", the theory is presented as being based on just two postulates:

1. Principle of Relativity: The laws of physics are invariant (identical) in all inertial frames of reference (that is, reference frames with no acceleration, that is, reference frames in which the Newton's laws are valid).
2. Principle of Light Constancy or Principle of Light Speed Invariance: The speed of light in vacuum is the same for all observers, regardless of the motion of light source or observer.

## II Terminology

- Speed of light $c$: the maximum speed of information, independent of the speed of the source and receiver.
- Clock: a device to measure differences in time.
- Event: something that happens at a definite place and time.
- Spacetime: geometrical space and time considered together.
- Spacetime interval between two events: a measure of separation between events that is defined as square their separation in time minus square their separation in geometrical space.
- Worldline: the path in spacetime of an observer.
- Reference frame or coordinate system: a way to locate events in spacetime. Events have coordinates $x, y, z$ for space and $t$ for time.
- Inertial reference frame or free-float reference frame: a reference frame where objects follow Newton's law.
- Coordinate transformation: changing how an event is described from one reference frame to another.
- Invariance: when physical laws or quantities do not change in different inertial frames. The speed of light is invariant in special relativity.

# III Lorentz transformation

## i Lorentz transformation and its inverse

Define an event to have spacetime coordinates $(t, x, y, z)$ in reference frame $S$ and $(t', x', y', z')$ in reference frame $S'$ moving at a velocity $v$ in positive $x$ direction. Then the Lorentz transformation specifies that these coordinates are related in the following way:

$$t' = \gamma\left(t - \frac{v}{c^2}x\right)$$

$$x' = \gamma(x - vt)$$

$$y' = y$$

$$z' = z$$

where

$$\gamma = \frac{1}{\sqrt{1-\beta^2}} = \frac{1}{\sqrt{1-\left(\frac{v}{c}\right)^2}}$$

is the Lorentz factor, $c$ is the speed of light, and $\beta = \dfrac{v}{c}$ is the speed parameter.

The inverse Lorentz transformation is obtained by solving the four Lorentz transformation equations as

$$t = \gamma\left(t' + \frac{v}{c^2}x\right)$$

$$x = \gamma(x' + vt')$$

$$y = y'$$

$$z = z'$$

A quantity that is invariant under Lorentz transformations is known as a Lorentz scalar.

Suppose one event has coordinates $(t_1, x_1, y_1, z_1)$ in $S$ and $(t'_1, x'_1, y_1, z_1)$ in $S'$ and another event has coordinates $(t_2, x_2, y_2, z_2)$ in $S$ and $(t'_2, x'_2, y_2, z_2)$ in $S'$. Let

$$\Delta t = t_2 - t_1$$

$$\Delta x = x_2 - x_1$$

$$\Delta t' = t'_2 - t'_1$$

$$\Delta x' = x'_2 - x'_1$$

We get

$$\Delta t' = \gamma\left(\Delta t - \frac{v}{c^2}\Delta x\right)$$

$$\Delta x' = \gamma(\Delta x - v\Delta t)$$

$$\Delta t = \gamma\left(\Delta t' + \frac{v}{c^2}\Delta x'\right)$$

$$\Delta x = \gamma(\Delta x' + v\Delta t')$$

If we take differentials instead of differences, we get

$$\mathrm{d}t' = \gamma\left(\mathrm{d}t - \frac{v}{c^2}\,\mathrm{d}x\right)$$

$$\mathrm{d}x' = \gamma(\mathrm{d}x - v\,\mathrm{d}t)$$
$$\mathrm{d}t = \gamma\left(\mathrm{d}t' + \frac{x}{c^2}\,\mathrm{d}x'\right)$$
$$\mathrm{d}x = \gamma(\mathrm{d}x' + v\,\mathrm{d}t')$$

### ii Derivation of Lorentz transformation

By the two postulates, we have

$$(ct)^2 - x^2 - y^2 - z^2 = (ct')^2 - x'^2 - y'^2 - z'^2.$$

Since the motion is along the $x$-axes only, we have

$$(ct)^2 - x^2 = (ct')^2 - x'^2.$$

Let

$$x' = f(x,t)$$
$$ct' = g(x,t)$$

Consider an object moving in a constant speed $u$ in $S$. It must be moving in a constant speed $w$ in $S'$.

$$f(ut,t) = \frac{w}{c}g(ut,t)$$
$$(c^2 - u^2)t^2 = af(ut,t)^2$$
$$bt = f(ut,t)$$

Thus $f$ must be linear with respect to $t$ and $\frac{\partial f}{\partial x}$ must not depend on $t$. Similarly, $f$ must be linear with respect to $x$ and $\frac{\partial f}{\partial t}$ must not depend on $x$.
That is, the solution must take the linear form

$$x' = Ax + Bct$$

$$ct' = Cx + Dct$$

Substitution gives

$$c^2t^2 - x^2 = C^2x^2 + 2CDcxt + D^2c^2t^2 - A^2x^2 - 2ABcxt - B^2c^2t^2$$

$$-1 = C^2 - A^2$$

$$A^2 - C^2 = 1$$

$$0 = 2CDcxt - 2ABcxt$$

$$AB = CD$$

$$c^2 = D^2c^2 - B^2c^2$$

$$D^2 - B^2 = 1$$

$$D = \frac{AB}{C}$$

$$A^2 - C^2 = \left(\frac{C}{B}\right)^2 = 1$$

Take

$$B = C$$

$$A = D \geq$$

Use

$$\cosh^2 \psi - \sinh^2 \psi = 1$$

Let

$$A = D = \cosh \psi$$

$$B = C = \sinh \psi$$

We have

$$x' = \cosh \psi x + \sinh \psi ct$$

$$ct' = \sinh \psi x + \cosh \psi ct$$

Substituting $x' = 0$ in $S'$ is measured as $x = vt$ in $S$

$$0 = \cosh \psi vt + \sinh \psi ct$$

$$ct' = \cosh \psi ct + \sinh \psi vt$$

Using identities

$$\sinh \psi = \frac{\tanh \psi}{\sqrt{1 - \tanh^2 \psi}}$$

$$\cosh \psi = \frac{1}{\sqrt{1 - \tanh^2 \psi}}$$

We have

$$0 = vt + \tanh \psi ct$$

$$\tanh \psi = -\frac{v}{c} = -\beta$$

$$\sinh \psi = \frac{-\beta}{\sqrt{1 - \beta^2}} = -\gamma \beta$$

$$\cosh \psi = \frac{1}{\sqrt{1 - \beta^2}} = \gamma$$

Substitution gives

$$x' = \gamma\,(x - vt)$$

$$ct' = \gamma(ct - \beta x)$$

$$t' = \gamma \left(t - \frac{v}{c^2} x\right)$$

# IV Consequence of Lorentz transformation

## i Invariant interval or Minkowski interval

In Galilean relativity, the spatial separation or separation in space $\sqrt{(\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2}$, and the temporal separation or separation in time, $\Delta t$, between two events are independent invariants.

In special relativity, however, they are not. Instead, the invariant interval, denoted as $s^2$ and defined as

$$s^2 = c^2(\Delta t)^2 - (\Delta x)^2 - (\Delta y)^2 - (\Delta z)^2,$$

is invariant.

*Proof.* Consider the Lorentz transformation setup.

$$s = c^2(\Delta t)^2 - (\Delta x)^2$$

$$\begin{aligned}
s'^2 =& c^2(\Delta t')^2 - (\Delta x')^2 \\
=& \gamma^2 \left((c\Delta t - \beta \Delta x)^2 - (\Delta x - v\Delta t)^2\right) \\
=& \gamma^2 \left(c^2(\Delta t)^2 - 2v\Delta x \Delta t + \beta^2(\Delta x)^2 - (\Delta x)^2 + 2v\Delta x\Delta t - v^2(\Delta t)^2\right) \\
=& \gamma^2 \left(\left(\frac{c}{\gamma}\right)^2 (\Delta t)^2 - \frac{1}{\gamma^2}(\Delta x)^2\right) \\
=& c^2(\Delta t)^2 - (\Delta x)^2 \\
=& s^2
\end{aligned}$$

□

There are three cases of $s^2$:

- $s^2 > 0$: This implies that $\left|\frac{\Delta x}{\Delta t}\right| < c$, called the two events are timelike separated. In this case, it is possible to find a inertial reference frame in which the two events happen at the same place. In this frame, the separation in time is $\frac{\sqrt{s^2}}{c}$, called proper time (interval).
- $s^2 < 0$: This implies that $\left|\frac{\Delta x}{\Delta t}\right| > c$, called the two events are spacelike separated. In this case, it is possible to find a inertial reference frame in which the two events happen at the same time. In this frame, the separation in space is $\sqrt{-s^2}$, called proper distance or proper length.
- $s^2 = 0$: This implies that $\left|\frac{\Delta x}{\Delta t}\right| = c$, called the two events are lightlike separated.

## ii Relativity of simultaneity

Two events that are simultaneous in frame $S$ (satisfying $\Delta t = 0$), are not necessarily simultaneous in another inertial frame $S'$ (satisfying $\Delta t' = 0$). Only if these events are additionally co-local in frame $S$ (satisfying $\Delta x = 0$), will they be simultaneous in another frame $S'$.

### iii Time dilation and Langevin's light-clock

Two events that happen at the same spatial point with time separation $\Delta t$ in $S$ has time separation $\Delta t'$ in $S'$:

$$\Delta t' = \frac{\Delta t}{\sqrt{1-\left(\frac{v}{c}\right)^2}} = \frac{\Delta t}{1+\beta^2} = \gamma \Delta t.$$

Langevin's light-clock:

A light-clock is imagined to be a box of perfectly reflecting walls wherein a light signal reflects back and forth from opposite faces.

Consider the scenario that observer A on a train holds a light-clock of $L$ as well as an electronic timer with which she measures how long it takes a pulse to make a round trip up and down along the light-clock. From her point of view the emission and receipt of the pulse occur at the same place, and she measures the interval using a single clock located at the precise position of these two events. For the interval between these two events, observer A finds

$$\Delta t_0 = \frac{2L}{c},$$

which is a proper time interval.

Another observer B, who is parked by the tracks as the train goes by at a speed of $v$. Instead of making straight up-and-down motions, observer B sees the pulses moving along a zig-zag line. However, because of the postulate of the constancy of the speed of light, the speed of the pulses along these diagonal lines is the same $c$. For the interval between these two events, observer B finds

$$\Delta t = \frac{2L}{\sqrt{c^2 - v^2}} = \frac{\Delta t_0}{\sqrt{1-\left(\frac{v}{c}\right)^2}},$$

which is not a proper time interval.

### iv Length contraction, Lorentz contraction, or Lorentz–FitzGerald contraction

The dimensions (e.g., length) of an object as measured by one observer may be smaller than the results of measurements of the same object made by another observer.

Suppose a measuring rod is at rest, aligned along the $x$-axis, and with length $L$ in $S$. To measure the length of this rod in $S'$, in which the rod is moving in velocity $-v$, the distance $L'$ must be measured simultaneously in $S'$, that is, $\Delta t' = 0$. We get

$$L' = \frac{L}{\gamma}.$$

### v Lorentz transformation of velocities or relativistic velocity addition or composition

Let

$$u = \frac{\mathrm{d}x}{\mathrm{d}t}$$

$$u' = \frac{\mathrm{d}x'}{\mathrm{d}t'}$$

Since

$$\mathrm{d}t' = \gamma\left(\mathrm{d}t - \frac{v}{c^2}\,\mathrm{d}x\right)$$
$$\mathrm{d}x' = \gamma(\mathrm{d}x - v\,\mathrm{d}t)$$

We have

$$\begin{aligned} u' =& \frac{\mathrm{d}x'}{\mathrm{d}t'} \\ =& \frac{\gamma(\mathrm{d}x - v\,\mathrm{d}t)}{\gamma\left(\mathrm{d}t - \frac{v}{c^2}\,\mathrm{d}x\right)} \\ =& \frac{\frac{\mathrm{d}x}{\mathrm{d}t} - v}{1 - \frac{v}{c^2}\frac{\mathrm{d}x}{\mathrm{d}t}} \\ =& \frac{u - v}{1 - \frac{uv}{c^2}} \end{aligned}$$

Check consistency

$$\begin{aligned} u' &= \frac{u - v}{1 - \frac{uv}{c^2}} \\ u' - \frac{u'uv}{c^2} &= u - v \\ u + \frac{uu'v}{c^2} &= u' + v \\ u &= \frac{u' + v}{1 + \frac{u'v}{c^2}} \end{aligned}$$

This replaces the $u' = u - v$ in Galilean relativity.

# V Four-vector, 4-vector, Lorentz vector, Minkowski space, or Minkowski spacetime

## i Four-vector, 4-vector, Lorentz vector, or contravariant four-tensor

A four-vector is of the form $A^\mu = (A^0, A^1, A^2, A^3)$.

Hereafter, $\mathbf{e}^0$ denotes the vector $(1, 0, 0, 0)$, $\mathbf{e}^1$ denotes the vector $(0, 1, 0, 0)$, $\mathbf{e}^2$ denotes the vector $(0, 0, 1, 0)$, $\mathbf{e}^3$ denotes the vector $(0, 0, 0, 1)$.

## ii Four-covector or covariant four-tensor

For a four-vector $A^\mu = (A^0, A^1, A^2, A^3)$, the corresponding four-covector is defined as either

$$A_\mu = (A^0, -A^1, -A^2, -A^3)$$

or

$$A_\mu = (-A^0, A^1, A^2, A^3)$$

in different conventions. We will hereafter call the former convention timelike convention and the latter spacelike convention.

Hereafter, $\mathbf{e}_0$ denotes the covector $(1, 0, 0, 0)$, $\mathbf{e}_1$ denotes the covector $(0, 1, 0, 0)$, $\mathbf{e}_2$ denotes the covector $(0, 0, 1, 0)$, $\mathbf{e}_3$ denotes the covector $(0, 0, 0, 1)$.

### iii Minkowski metric tensor, Minkowski tensor, Minkowski metric, or metric

The Minkowski metric tensor, Minkowski tensor, Minkowski metric, or simply metric, denoted as $\eta$, is defined such that

$$\eta_{\mu\nu}A^{\mu} = A_{\nu}$$

$$\eta^{\mu\nu}A_{\mu} = A^{\nu}.$$

For timelike convention, that is,

$$\eta = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix},$$

which has signature $(+,-,-,-)$.

For spacelike convention, that is,

$$\eta = \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix},$$

which has signature $(-,+,+,+)$.

### iv Lorentz transformation matrix

The Lorentz transformation from system $S$ with coordinate $x^{\mu}$ and to system $S'$ with coordinate $x'^{\mu}$ is

$$x'^{\mu} = \Lambda^{\mu}_{\nu}x^{\nu}$$

where

$$\Lambda = \begin{bmatrix} \gamma & -\beta\gamma & 0 & 0 \\ -\beta\gamma & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

This $\Lambda$ is called a Lorentz transformation matrix.

For any Lorentz transformation matrix $\Lambda$, we have, in matrix notation

$$\Lambda^{\top}\eta\Lambda = \eta$$

or equivalently, in tensor notation

$$\eta_{\mu\nu}\Lambda^{\mu}_{\rho}\Lambda^{\nu}_{\sigma} = \eta_{\rho\sigma}.$$

*Proof.* Timelike convention:

$$\begin{bmatrix} \gamma & -\beta\gamma & 0 & 0 \\ -\beta\gamma & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}\begin{bmatrix} \gamma & -\beta\gamma & 0 & 0 \\ -\beta\gamma & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} \gamma & \beta\gamma & 0 & 0 \\ -\beta\gamma & -\gamma & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}\begin{bmatrix} \gamma & -\beta\gamma & 0 & 0 \\ -\beta\gamma & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} (1-\beta^2)\gamma^2 & 0 & 0 & 0 \\ 0 & -(1-\beta^2)\gamma^2 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$$

Spacelike convention: Timelike convention multiplied by $-1$. □

## v Minkowski dot product or Lorentz-invariant inner product

The Minkowski dot product or Lorentz-invariant inner product of two four-vectors is defined as

$$A \cdot B := A^\mu \eta_{\mu\nu} B^\nu.$$

A Minkowski dot product is invariant under Lorentz transformations, that is,

$$A^\mu \eta_{\mu\nu} B^\nu = (\Lambda^\mu_\rho A^\rho)\eta_{\mu\nu}(\Lambda^\nu_\sigma B^\sigma).$$

*Proof.* Rearrange RHS.

$$\eta_{\mu\nu}\Lambda^\mu_\rho \Lambda^\nu_\sigma = \eta_{\rho\sigma}.$$

$$(\Lambda^\mu_\rho A^\rho)\eta_{\mu\nu}(\Lambda^\nu_\sigma B^\sigma) = A^\rho \eta_{\rho\sigma} B^\sigma = A^\mu \eta_{\mu\nu} B^\nu.$$

□

## vi Minkowski invariant

For a four-vector $A^\mu = (A^0, A^1, A^2, A^3)$, the Minkowski invariant is defined as $A^\mu A_\mu$, which equals to $(A^0)^2 - (A^1)^2 - (A^2)^2 - (A^3)^2$ for timelike convention and $-(A^0)^2 + (A^1)^2 + (A^2)^2 + (A^3)^2$ for spacelike convention.

This quantity is invariant under Lorentz transformations.

*Proof.*

$$A'^\mu = \Lambda^\mu_\nu A^\nu$$

We want to show that

$$A'^{\mu} A'_{\mu} = A^{\mu} A_{\mu}.$$

$$A'_{\mu} = \eta_{\mu\rho} A'^{\rho} = \eta_{\mu\rho} \Lambda^{\rho}_{\nu} A^{\nu}$$

$$\begin{aligned} A'^{\mu} A'_{\mu} =& \left(\Lambda^{\mu}_{\alpha} A^{\alpha}\right) \left(\eta_{\mu\rho} \Lambda^{\rho}_{\beta} A^{\beta}\right) \\ =& A^{\alpha} A^{\beta} \left(\Lambda^{\mu}_{\alpha} \eta_{\mu\rho} \Lambda^{\rho}_{\beta}\right) \\ =& A^{\alpha} A^{\beta} \eta_{\alpha\beta} \\ =& A^{\alpha} A_{\alpha} \\ =& A^{\mu} A_{\mu} \end{aligned}$$

□

### vii Differential four-vector

The differential of a four-vector $A^{\mu}$ is defined as

$$\mathrm{d}A^{\mu} = \left(\mathrm{d}A^0, \mathrm{d}A^1, \mathrm{d}A^2, \mathrm{d}A^3\right).$$

### viii Position four-vector, four-position, or 4-position

In Newtonian physics, we use three-position

$$\mathbf{r} = (x, y, z)$$

to represent space.

In special relativity, we combine space and time into a single object

$$x^{\mu} = (ct, x, y, z),$$

called position four-vector, four-position, or 4-position.

### ix Displacement four-vector or four-displacement

The displacement four-vector is defined as

$$\Delta x^{\mu} = (c\Delta t, \Delta x, \Delta y, \Delta z).$$

### x Proper time differential

Proper time differential $\mathrm{d}\tau$ is defined as

$$\mathrm{d}\tau = \frac{1}{c}\sqrt{c^2(\mathrm{d}t)^2 - (\mathrm{d}x)^2 - (\mathrm{d}y)^2 - (\mathrm{d}z)^2} = \sqrt{(\mathrm{d}t)^2 - \frac{(\mathrm{d}x)^2 + (\mathrm{d}y)^2 + (\mathrm{d}z)^2}{c^2}}.$$

### xi Velocity four-vector or four-velocity

The velocity four-vector is defined as

$$u^{\mu} = \frac{\mathrm{d}x^{\mu}}{\mathrm{d}\tau}.$$

## xii Acceleration four-vector or four-acceleration

The acceleration four-vector is defined as

$$a^\mu = \frac{\mathrm{d}u^\mu}{\mathrm{d}\tau}.$$

## xiii Four-gradient

In timelike convention, the four-gradient for vector $\partial^\mu$ is defined as

$$\begin{aligned}\partial^\mu =& \left(\frac{\partial}{\partial x_0}, -\frac{\partial}{\partial x_1}, -\frac{\partial}{\partial x_2}, -\frac{\partial}{\partial x_3}\right)\\ =&\mathbf{e}_0\frac{1}{c}\frac{\partial}{\partial t} - \nabla\end{aligned}$$

the four-gradient for covector $\partial_\mu$ is defined as

$$\begin{aligned}\partial_\mu =& \left(\frac{\partial}{\partial x^0}, \frac{\partial}{\partial x^1}, \frac{\partial}{\partial x^2}, \frac{\partial}{\partial x^3}\right)\\ =&\mathbf{e}^0\frac{1}{c}\frac{\partial}{\partial t} + \nabla\end{aligned}$$

and

$$\partial^\mu\partial_\mu = \frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \nabla^2.$$

In spacelike convention, the four-gradient for vector $\partial^\mu$ is defined as

$$\begin{aligned}\partial^\mu =& \left(-\frac{\partial}{\partial x_0}, \frac{\partial}{\partial x_1}, \frac{\partial}{\partial x_2}, \frac{\partial}{\partial x_3}\right)\\ =& -\mathbf{e}_0\frac{1}{c}\frac{\partial}{\partial t} + \nabla\end{aligned}$$

the four-gradient for covector $\partial_\mu$ is defined as

$$\begin{aligned}\partial_\mu =& \left(\frac{\partial}{\partial x^0}, \frac{\partial}{\partial x^1}, \frac{\partial}{\partial x^2}, \frac{\partial}{\partial x^3}\right)\\ =&\mathbf{e}^0\frac{1}{c}\frac{\partial}{\partial t} + \nabla\end{aligned}$$

and

$$\partial^\mu\partial_\mu = -\frac{1}{c^2}\frac{\partial^2}{\partial t^2} + \nabla^2.$$

## xiv Wave four-vector

For a plane wave with angular frequency $\omega$ and wave vector $\mathbf{k}$, the wave four-vector $k^\mu$ is defined as

$$k^\mu = \left(\frac{\omega}{c}, \mathbf{k}\right).$$

For light in vacuum, that is,

$$k^\mu = \frac{\omega}{c}\left(1, \hat{\mathbf{k}}\right).$$

## VI Relativistic Doppler Effect in Vacuum

Let the unit vector in the direction from the source to the receiver be $\hat{\mathbf{n}}$, $c$ be the speed of light, $v$ be the speed of the receiver relative to the source, $\beta = \frac{v}{c}$, $\theta$ be the polar angle of the velocity of the receiver relative to the source in the polar coordinate system on the common plane of the velocity of the receiver relative to the source and $\hat{\mathbf{n}}$ with $\hat{\mathbf{n}}$ as $\theta = 0$, $f$ and $\lambda$ be the frequency and wavelength of the light as observed by the source, and $f'$ and $\lambda'$ be the frequency and wavelength of the light as observed by the receiver. Relativistic Doppler effect states that

$$\lambda = \frac{c}{f}$$

$$f' = \frac{\sqrt{1-\beta^2}}{1-\beta\cos\theta} f = \frac{1}{\gamma(1-\beta\cos\theta)} f$$

$$\lambda' = \frac{c}{f'} = \frac{1-\beta\cos\theta}{\sqrt{1-\beta^2}}\lambda = \gamma(1-\beta\cos\theta)\lambda$$

*Proof.* The four-velocity of the receiver relative to the source is

$$u^\mu = \gamma(c, v\cos\theta, v\sin\theta\cos\varphi, v\sin\theta\sin\varphi)$$

Use timelike convention,

$$\begin{aligned}\omega' &= k_\mu u^\mu \\ &= \frac{\omega}{c}\gamma(c - v\cos\theta) \\ &= \omega\gamma(1-\beta\cos\theta)\end{aligned}$$

□

For $\theta = 0$,

$$f' = \sqrt{\frac{1+\beta}{1-\beta}} f$$

$$\lambda' = \sqrt{\frac{1-\beta}{1+\beta}}\lambda$$

For $\theta = \frac{\pi}{2}$,

$$f' = \sqrt{1-\beta^2} f = \frac{1}{\gamma} f$$

$$\lambda' = \frac{1}{\sqrt{1-\beta^2}}\lambda = \gamma\lambda$$

## VII Momentum and energy

PLACEHOLDER prove with Noether's theorem and its proof

### i Relativistic momentum

The relativistic momentum's is defined as

$$\mathbf{p} = \gamma m \mathbf{v}$$

where $m$ is mass, $\mathbf{v}$ is velocity, $v = |\mathbf{v}|$, and $\gamma = \dfrac{1}{\sqrt{1 - \left(\frac{v}{c}\right)^2}}$.

For $v \ll c$, $\gamma \approx 1$, $\mathbf{p} \approx m\mathbf{v}$.

As $v \to c$, $\gamma \to \infty$, thus a massive particle cannot be accelerated to $c$ with finite energy.

### ii Relativistic energy

The relativistic energy is defined as

$$E = \gamma m c^2.$$

Thus when $v = 0$, $E = mc^2$. This is called mass–energy equivalence.

We call $E$ (total) energy, $E_0 = mc^2$ (total) mass/rest energy, and $K = E - mc^2 = mc^2(\gamma - 1)$ kinetic energy.

For $v \ll c$, the first-order approximation of $\dfrac{1}{\sqrt{1-x}}$ at $0$ is

$$\frac{1}{\sqrt{1-x}} \approx 1 + \frac{x}{2}$$

thus

$$\gamma \approx 1 + \frac{v^2}{2c^2}$$

$$K = mc^2(\gamma - 1) \approx \frac{mv^2}{2}$$

Let $p = \left|\mathbf{p}\right|$ we have

$$(pc)^2 = K^2 + 2Kmc^2$$

$$E^2 = (pc)^2 + (mc^2)^2$$

*Proof.*

$$\begin{aligned}
K^2 + 2Kmc^2 &= \left((\gamma - 1)^2 + 2(\gamma - 1)\right)(mc^2)^2 \\
&= \left(\gamma^2 - 1\right)(mc^2)^2 \\
&= \left(\frac{1}{1 - \left(\frac{v}{c}\right)^2} - 1\right)(mc^2)^2 \\
&= \frac{(mvc)^2}{1 - \left(\frac{v}{c}\right)^2} \\
&= (\gamma m v c)^2 \\
&= (pc)^2
\end{aligned}$$

$$
\begin{aligned}
(pc)^2 + (mc^2)^2 &= \left((\gamma v)^2 + c^2\right)(mc)^2 \\
&= \left( \frac{v^2}{1 - \left(\frac{v}{c}\right)^2} + c^2 \right)(mc)^2 \\
&= \left( \frac{c^2}{1 - \left(\frac{v}{c}\right)^2} \right)(mc)^2 \\
&= (\gamma mc^2)^2
\end{aligned}
$$

□

### iii Four-momentum

The four-momentum's is defined as

$$p^\mu = mu^\mu,$$

that is,

$$p^\mu = (\gamma mc, \gamma m\mathbf{v}) = \left(\frac{E}{c}, \mathbf{p}\right).$$

The four-momentum is conservative.

### iv Q value

In a system undergoing a chemical or nuclear reaction, a change in the total kass energy of the system due to the reaction is often given as $Q$ value, defined as the system's initial total mass energy $E_{0i}$ minus the system's final total mass energy $E_{0f}$, that is,

$$E_{0i} = E_{0f} + Q.$$

We can also write in terms of the initial total mass $M_i$ and the final total mass $M_f$ as

$$M_i c^2 = M_f c^2 + Q = -\Delta M c^2,$$

where $\Delta K = M_f - M_i$ is the change in mass due to the reaction.